Cristhian Kevin Mallqui Castro

Robot Learning | Perception | Autonomous Systems
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Profile

MSc student and Research Engineer at KTH interested in how robots learn, understand their surroundings and act autonomously in the real world. My path from mechatronics in Peru to international robotics research combines hands-on engineering with a commitment to collaboration, teaching and mentorship.

Education

KTH Royal Institute of Technology *Aug 2025 – Present*
MSc in Systems, Control and Robotics | Stockholm, Sweden
Autonomous Systems specialization; expected completion in 2027. Study includes robotics, modelling, computer vision, applied estimation, machine learning and embedded systems.

Pontificia Universidad Católica del Perú *Mar 2018 – Dec 2023*
BSc in Mechatronics Engineering | Lima, Peru
Top 10% of graduating class. **Thesis:** Design of a teleoperated vehicle for weld-seam inspection and corrosion mapping on hard-to-reach ferromagnetic surfaces. Supervisor: Prof. Diego Quiroz.

Lund University *Jan – Jun 2023*
Exchange studies in Mechatronics | Lund, Sweden
Projects in real-time control systems and mechatronic industrial product design.

Research Experience

Research Engineer | KTH Robotics, Perception and Learning *Oct 2025 – Present*
ALARS project | Supervisors: Prof. John Folkesson and Özer Özkahraman, PhD

- Develop perception for autonomous aerial recovery of underwater robots; calibrate a 360-degree camera and train YOLO detectors using real and synthetic imagery.
- Work on a DJI Matrice 350 landing pipeline that connects oriented object detections, EKF relative-state estimation, a landing state machine and LQR control in ROS 2.
- Develop an automated labeling pipeline and frame selection based on VIsual Foundation Models.

Summer Research Assistant | University of Delaware *Jul – Nov 2024*
Soft robotics | Supervisor: Prof. Nathan Lazarus

- Led two projects on liquid-metal vibrotactile actuators and magnetic adhesion systems, combining prototype design, fabrication, performance characterization and safety analysis.
- Validated wearable prototypes through user testing; research resulted in a first-author journal publication and an invention disclosure in progress.

Research Intern | PUCP Core Facilities – FABCORE *Aug 2023 – Feb 2024*
Designed and implemented a 1.5 m clay delta 3D printer across mechanics, electronics, software and control; trained new interns and supported 3D-printing services.

Projects

Mobile Manipulation and Imitation Learning | RBY1 / MoBiPi *Ongoing*
Supervisor: Lakshadeep Surendra Naik, Postdoctoral Researcher | KTH

- Develop MuJoCo simulation and demonstration pipelines for RBY1 manipulation using robosuite, RoboCasa, robomimic and MimicGen; train and evaluate Diffusion Policy.
- Validate an action adapter across 100 simulator states, including contact configurations; investigate controller representations, demonstration replay and policy rollout behavior.

RGB Room Transitions and Topological Scene Representation *Ongoing*
Supervisors: Dr. Olov Andersson, Timon Homberger, PhD, and Finn Busch, PhD

- Develop room-transition detection with RADIO embeddings and temporal convolutional networks; evaluate on Spot/Odin indoor recordings and collect RGB trajectories in Habitat-Sim with HM3D.
- Investigate room-transition cues as a component of localization and mapping pipelines.

Projects continued

Vehicle State Estimation | KTH Formula Student *Oct 2025 – Jan 2026*

Supervisor: Prof. John Folkesson

- Developed and tested a cascaded UKF–EKF for velocity and attitude estimation under wheel slip, using adaptive filtering and innovation gating on real vehicle data at 100 Hz.

Illary Rover | European Rover Challenge *Feb – Aug 2025*

Supervisor: Dr. Quino Valverde

- With GRIP, implemented multi-mode steering and ROS 2 autonomy on Jetson Orin NX, integrating RealSense D455, Livox MID-360, visual SLAM, sensor fusion and nvblox costmaps.

Publications

Liquid Touch: Soft and Wearable Vibrotactile Actuators Based on Room Temperature Liquid Metals. *Advanced Engineering Materials*, 2026. First-author journal paper. Invention disclosure in process.

Gone Fishing: Post-Mission Autonomous Aerial Recovery of Unmanned Underwater Vehicles. *IEEE OES AUV Symposium*, 2026. Accepted extended abstract; contribution to underwater perception for ALARS.

Robotics Summer Schools

ETH RobotX Robotics Summer School *2026*

- Tuned exploration and state-estimation parameters and worked on object-detection pipelines with persistent detections in the provided ROS 2 autonomy stack.
- Tested the pipeline in MuJoCo and transferred it to a real Unitree A2 quadruped.

RimA Summer School | FZI and KIT *3–7 Aug 2026*

- Developed shoe detection and classification, including recognition of unknown shoes, using vision foundation models and YOLO for the Unitree Go2.
- Implemented behavior trees on Boston Dynamics Spot for navigation and manipulation tasks in indoor environments.

Teaching and Leadership

University Teaching Assistant | PUCP *Mar 2024 – Aug 2025*

Taught robotics, AI, mechatronic design and physics, including kinematics and robot motion planning.

Robotics Teacher | UTK / Toulouse Lautrec *Jan – May 2025*

Taught robotics and AI to children; guided projects combining programming, electronics and mechanical design, and supported STEM laboratory activities.

Leadership and Mentoring | PUCP Robotics Group *Nov 2018 – Aug 2025*

Served as President, Vice President and Workshop Coordinator; mentored robotics competition teams.

Skills

Programming and development: Python, C/C++, MATLAB, Linux, Git, Docke.
Robotics and estimation: ROS 2, SLAM, Nav2, EKF/UKF sensor fusion, MoveIt 2.
Computer vision and AI: OpenCV, PyTorch, Foundation Models, imitation learning, Diffusion Policy.
Simulation and data: MuJoCo, MimicGen, RoboCasa, Habitat-Sim, Isaac Sim, Unity.
Sensors and hardware: RealSense RGB-D cameras, LiDAR, Jetson, Raspberry Pi, STM32.
Mechanical design: SolidWorks, Fusion 360, Inventor; robotic prototyping and system integration.
Languages: Spanish (native); English.

Team Awards

Latin American Space Challenge 2023 – CanSat first place; Satellite Challenge overall winner. Developed and integrated a transformable-wheel rover, including prototyping and power–sensor integration. Supervisor: Prof. Diego Arce

Latin American Space Challenge 2021 – CanSat first place. Contributed to an autogyro-based deployment system for agricultural applications, including system integration and validation. Supervisor: Prof. Diego Arce

European Rover Challenge Remote 2023 – First in Latin America; 11th overall. Development of autonomous solution for marker and panel inspection. Supervisor: Prof. Diego Arce